

A Cloud-Edge Hybrid Intelligent Scheduling Method for Pumping Stations Integrating RAG and Multi-Agent Collaboration

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Abstract—In the face of extreme weather-induced urban waterlogging and the inefficiency of traditional water conservancy scheduling, existing approaches based on static rules or model-free deep reinforcement learning exhibit significant limitations in dynamic environmental responsiveness and trial-and-error costs. To address these challenges, this paper proposes a novel cloud-edge hybrid intelligent scheduling architecture that synergizes Large Language Models (LLMs) with Multi-Agent systems. At the infrastructure level, the system is built upon a FastAPI-based proxy, integrating the Model Context Protocol (MCP) for dynamic tool execution and Retrieval-Augmented Generation (RAG) for domain-specific knowledge enhancement [4]. Operationally, grounded in digital twin concepts and game theory, we design a six-step dynamic regulation closed-loop workflow encompassing requirement identification, state assessment, hydraulic calculation, and command dispatch [11]. This research aims to provide a comprehensive theoretical framework and a practical engineering paradigm for the intelligent, adaptive, and energy-efficient management of large-scale pumping station clusters. Extensive simulations demonstrate that the proposed method significantly outperforms traditional heuristic and standard MARL baselines, reducing lifecycle energy consumption by 12.5%, decreasing limit-exceeding alarms by over 80%, and providing 2 to 4 hours of preventative action lead time during extreme weather events.

Keywords: Large Language Models (LLMs), Multi-Agent Systems, Cloud-Edge Collaboration, Flood Control Scheduling, Reinforcement Learning, Retrieval-Augmented Generation (RAG).

I. INTRODUCTION

A. Research Background and Significance

In the context of intensifying global climate change and rapid urbanization, the frequent occurrence of extreme rainstorms has significantly elevated the risk of urban waterlogging. Large-scale pumping station clusters serve as the core infrastructure and the “heart” of urban flood control and drainage systems. Furthermore, under the macroscopic “Dual Carbon” (carbon peak and carbon neutrality) goals,

optimizing the energy efficiency of these massive hydraulic hubs presents a critical challenge. Traditional scheduling methods, which heavily rely on manual experience and rigid threshold control, often struggle to adapt to highly dynamic hydro-meteorological environments and complex boundary conditions. Consequently, these approaches frequently result in severe operational inefficiencies, including high no-load operation rates and significant response lags during emergency flood control scenarios.

B. Evolution and Bottlenecks of Existing Technologies

To address the inefficiencies of traditional scheduling, researchers have explored various technological paradigms, yet significant bottlenecks remain. Early data-driven approaches, which combine Internet of Things (IoT) platforms with traditional machine learning models, are inherently constrained by the complexity of multi-modal and heterogeneous data, leading to limited prediction accuracy and poor generalization in dynamic environments. More recently, Deep Reinforcement Learning (DRL) has been introduced to achieve optimal control in hydraulic engineering [2], [3], [9], [16]. However, DRL algorithms face severe challenges when applied to high-stakes physical domains. Specifically, they suffer from low sample efficiency, immense difficulties in practical deployment due to prohibitive trial-and-error costs, and slow convergence when subjected to strict, real-world safety constraints.

C. Main Contributions

To resolve the persistent disconnect between semantic cognition and physical execution, and to overcome the limitations of traditional DRL, this study proposes a novel cloud-edge hybrid intelligent scheduling framework. The main contributions of this paper are summarized as follows:

- **Hierarchical Collaborative Architecture:** We propose a hierarchical intelligent scheduling architecture that synergizes Large Language Models (LLMs) with Multi-Agent

systems, specifically tailored for the complex coordination of large-scale pumping station clusters.

- **RAG-Enhanced Global Control Expert System:** We construct a global control expert system enhanced by Retrieval-Augmented Generation (RAG). By integrating a domain-specific water conservancy knowledge base (including physical equations and historical scheduling plans), we provide *a priori* knowledge to support the LLM, effectively suppressing “hallucinations” in vertical domain applications.
- **DP-RL Hybrid Adaptive Scheduling Mechanism:** Incorporating multi-dimensional constraints, we design a single-station game and adaptive scheduling mechanism utilizing a hybrid fine-tuning optimization of Dynamic Programming (DP) and Reinforcement Learning (RL). This ensures both a deterministic safety boundary and highly dynamic responsiveness.

The remainder of this paper is organized as follows: Section II introduces the proposed cloud-edge collaborative system architecture. Section III details the core methodology, including the multi-modal data fusion, DP-RL hybrid optimization, and the LLM-driven global coordination. Section IV presents the experimental setup, simulation results, and quantitative performance analysis. Finally, Section V concludes the paper and discusses future work.

II. SYSTEM ARCHITECTURE DESIGN

To effectively manage the complexity of large-scale pumping station clusters and overcome the limitations of centralized control, this study designs a hierarchical architecture, as illustrated in Fig. 1.

A. Overall Architecture: Cloud-Edge Collaborative Model

The proposed system adopts a Cloud-Edge-Device collaborative topology, primarily divided into three functional layers to ensure both macroscopic optimality and microscopic responsiveness:

- **Data Acquisition and Edge Perception Layer:** This foundational layer is responsible for the real-time ingestion of multi-modal heterogeneous data. It seamlessly integrates continuous IoT sensor streams—such as water level, electrical current, and equipment vibration—with external meteorological APIs, constructing a comprehensive and dynamic environmental awareness state.
- **Local Agent Layer (Edge):** Deployed at the edge near the physical infrastructure, this layer focuses on the dynamic identification of single-station equipment statuses, short-term energy efficiency evaluations, and low-latency local control. It guarantees immediate, robust responsiveness to sudden hydraulic changes while strictly maintaining fundamental equipment safety constraints.
- **Global Coordination Layer (Cloud):** Acting as the central cognitive brain, the cloud layer utilizes Large Language Models (LLMs) to perform macro-level task decomposition, global state coordination, and cross-regional

resource scheduling among multiple distributed pumping stations [13], [18], [17].

B. LLM-Driven RAG Domain Knowledge Fusion

While general LLMs exhibit powerful reasoning capabilities, their lack of domain-specific hydraulic knowledge often leads to erroneous decision-making (i.e., “hallucinations”). To mitigate this, we propose an LLM-driven knowledge fusion mechanism based on Retrieval-Augmented Generation (RAG) [4].

- **Vector Knowledge Base Construction:** We construct a high-dimensional vector database embedding specialized water conservancy knowledge. This robust repository encompasses fundamental hydraulic physical equations (e.g., head loss, Manning’s equation), historical scheduling experience logs, and strict emergency safety protocols.
- **Decision Correction via RAG-Based Prompts:** During the dynamic inference phase, the system retrieves relevant physical constraints and historical precedents based on the real-time hydro-meteorological context. These retrieved semantics are fused into optimized prompts, effectively correcting the LLM’s cognitive biases and ensuring that generated scheduling strategies strictly adhere to physical reality and engineering safety regulations.

C. Agent Communication and Game Network

The collaborative execution of global scheduling strategies relies on a robust communication topology and a rigorous multi-agent negotiation mechanism [14], [15].

- **Lightweight Communication Topology:** To ensure real-time and reliable data transmission between the cloud coordinator and edge agents, the system adopts a hybrid, lightweight communication topology. It utilizes MQTT for low-bandwidth telemetry data reporting and gRPC for high-performance, low-latency execution commands.
- **Nash Equilibrium-Based Game Mechanism:** Resource allocation across multiple pumping stations is formulated as a cooperative-competitive game. By introducing a multi-agent game network based on the Nash equilibrium [1], the system dynamically negotiates the start-stop sequences and flow allocation quotas among different stations. This mechanism mathematically balances individual station energy efficiency with global flood control objectives, preventing suboptimal heuristic decisions.

III. KEY TECHNOLOGIES AND METHODOLOGY

This section details the core algorithmic framework of the proposed adaptive scheduling system, which encompasses multi-modal data fusion, hybrid local decision optimization, and global LLM-driven coordination.

A. Multi-modal Data Feature Fusion and State Representation

Accurate state representation is the prerequisite for intelligent scheduling. The system operates in a highly dynamic environment governed by both external meteorological factors

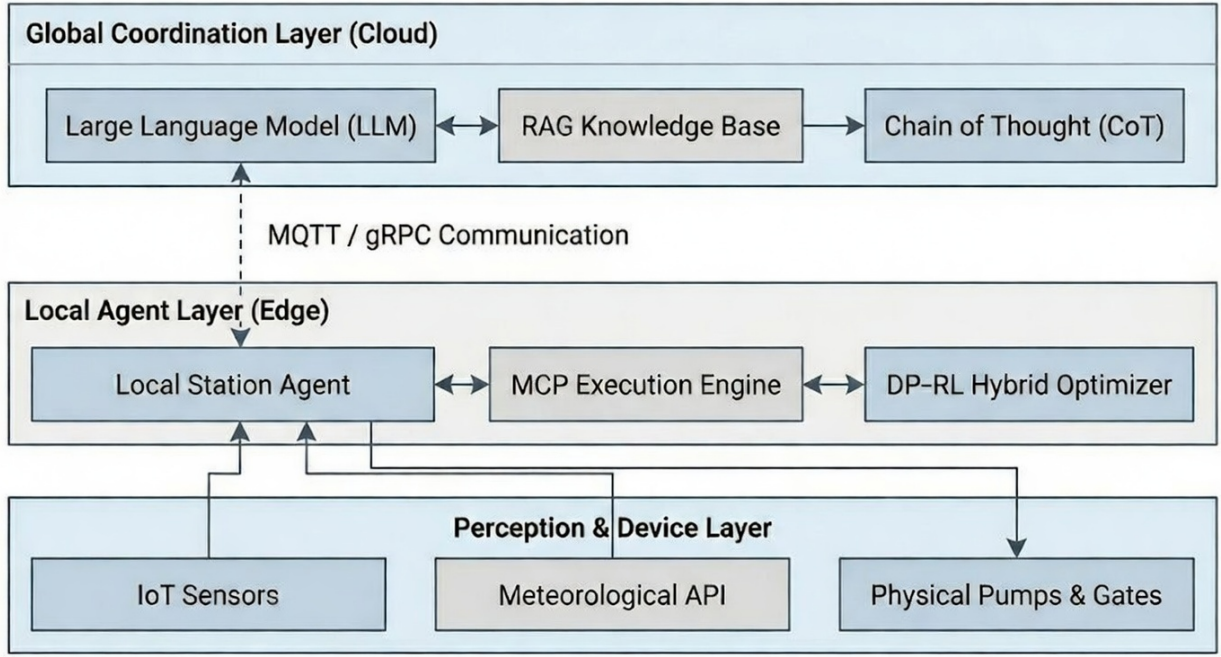


Fig. 1. The proposed cloud-edge hybrid intelligent scheduling architecture synergizing RAG-enhanced LLMs and collaborative multi-agents.

and internal electromechanical constraints. To construct a comprehensive state representation, we propose a joint time-series feature extraction module.

The overall state space at time step t , denoted as S_t , is defined by fusing environmental parameters and physical equipment states:

$$S_t = \{E_t, P_t\} \quad (1)$$

where the environmental state $E_t = [h_t^{(1)}, \dots, h_t^{(m)}, r_t, \dots, r_{t+k}]$ includes multi-section water levels h and short-term rainfall forecasts r over a horizon k . The physical state $P_t = [u_t^{(1)}, \dots, u_t^{(n)}, I_t^{(1)}, \dots, I_t^{(n)}]$ represents the operational status $u \in \{0, 1\}$ and electrical currents I of n pump units.

To effectively capture the temporal dependencies and spatial correlations within this heterogeneous data, we employ sequence-to-sequence deep learning architectures (e.g., LSTM or Transformer-based encoders) to map the raw multi-modal inputs into a high-dimensional dense vector representation z_t , which subsequently serves as the standardized input for the downstream decision-making agents.

B. DP-RL based Local Agent Decision Optimization

Traditional reinforcement learning often struggles with slow convergence and high safety risks in physical environments. To address this, we design a local agent decision optimization mechanism based on a Dynamic Programming and Reinforcement Learning (DP-RL) hybrid framework.

Baseline Strategy (DP): Under normal and predictable operating conditions, the system utilizes DP to generate a deterministic, optimal start-stop sequence. By incorporating the Time-of-Use (ToU) electricity pricing c_t and the basic

drainage load demand D_t , the DP algorithm solves for a baseline action a_t^{DP} that minimizes the operational cost while strictly adhering to safety bounds:

$$V^*(S_t) = \min_{a_t^{DP}} (c_t \cdot P(a_t^{DP}) + \gamma V^*(S_{t+1})) \quad (2)$$

subject to $h_{min} \leq h_{t+1} \leq h_{max}$, ensuring a robust and fault-tolerant foundational strategy.

Dynamic Fine-tuning (RL): For non-deterministic scenarios such as sudden rainstorms or unit failures, a proximal RL network (e.g., Proximal Policy Optimization, PPO) is activated. The RL agent outputs a residual continuous action space fine-tuning Δa_t^{RL} to compensate for the DP baseline. The final executed action is formulated as:

$$a_t = a_t^{DP} + \Delta a_t^{RL} \quad (3)$$

This hybrid approach guarantees fundamental safety limits via DP while preserving the high-dynamic adaptability of RL for real-time adjustments.

C. Global LLM Commander Multi-objective Coordination Algorithm

While local agents optimize individual station performance, the Global LLM Commander handles macro-level task decomposition and cross-regional resource scheduling.

To process complex hydro-meteorological conditions, the LLM utilizes a Chain of Thought (CoT) prompting strategy [5]. The CoT guides the LLM to logically deduce the situation step-by-step: assessing target capacity, verifying physical constraints, and formulating sequential coordination plans.

The global coordination is fundamentally a multi-objective optimization problem. The LLM acts as an expert heuristic

solver to approximate the Pareto optimal allocation among three conflicting objectives: 1) Strict adherence to safety water level limits, 2) Minimization of overall energy consumption, and 3) Minimization of equipment lifecycle wear caused by frequent start-stop cycles. The multi-objective function J can be mathematically represented as:

$$\min J = [f_{\text{energy}}(a), f_{\text{wear}}(a), f_{\text{safety_penalty}}(S)] \quad (4)$$

By leveraging RAG-enhanced domain knowledge, the LLM evaluates the trade-offs on the Pareto front, ensuring that emergency flood control targets are met without causing excessive mechanical degradation or unnecessary power grid burden.

D. Multi-Agent Six-Step Closed-Loop Scheduling Workflow

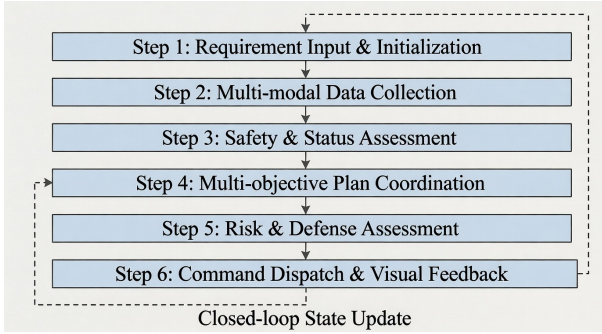


Fig. 2. The six-step dynamic regulation closed-loop workflow of the multi-agent system.

The core methodology is operationalized through a complex, six-stage intelligent agent scheduling workflow, as shown in Figure 2.

- 1) **Requirement Input and Initialization:** The system perceives multi-objective tasks, such as specific water transfer targets within a defined time window.
- 2) **Multi-modal Data Collection:** The agents acquire multi-source constraint data, including real-time flow rates, gate openings, unit switch states, and future maintenance schedules.
- 3) **Safety and Current Status Assessment:** The system calculates target gaps and evaluates hydraulic feasibility, dynamically adjusting water level predictions and verifying equipment maintenance conflicts.
- 4) **Multi-objective Plan Coordination:** Based on RAG retrieval and Nash equilibrium principles, the system automatically derives phased start-stop plans. It defines precise operational sequences, such as "15-hour preparation filling → operational mode switching → progressive target achievement".
- 5) **Risk and Defense Assessment:** Prior to execution, a secondary check is performed to evaluate equipment failure probabilities and water level boundary limits. The system seamlessly integrates flood warnings with corresponding mitigation measures, generating fault-tolerant plans such as backup unit switching.

- 6) **Documentation and Visual Self-feedback:** Finally, the derived plan is converted into standard Markdown operational commands. A backend LLM dynamically maps these commands to a dark-themed situational awareness dashboard equipped with ECharts views, facilitating human-machine collaborative verification.

IV. EXPERIMENTS AND RESULTS

A. Experimental Setup and Simulation Environment

To validate the effectiveness of the proposed cloud-edge hybrid scheduling architecture, we constructed a high-fidelity digital twin simulation environment [11]. The physical hydraulic dynamics were simulated using the Storm Water Management Model (SWMM) [6] coupled with a customized OpenAI Gym interface [7], enabling seamless interaction between the multi-agent system and the simulated pumping station cluster. Multi-modal input data, including historical rainfall, multi-section water levels, and equipment constraints, were injected into the environment to simulate real-world extreme weather scenarios.

B. Baseline Methods and Evaluation Metrics

We compared the proposed LLM-driven DP-RL framework against two representative baselines. Related pump scheduling studies are also reported in recent literature [10], [11], [12]:

- **Traditional Heuristic Rule-based Control (THRC):** The conventional scheduling strategy that heavily relies on static water level thresholds and manual dispatching experience.
- **Standard Multi-Agent PPO (MAPPO):** A conventional multi-agent proximal policy optimization (MAPPO) approach without the guidance of the LLM global coordinator or RAG domain knowledge enhancement.

The performance of the models was evaluated across two primary dimensions: Energy Efficiency and Operational Safety. Furthermore, the Nash-Sutcliffe Efficiency (NSE) coefficient [8] was utilized alongside standard metric functions to assess the predictive accuracy of the hydraulic state representation module under dynamic flood routing. The overall quantitative comparison is shown in Fig. 3.

C. Results Analysis

TABLE I
QUANTITATIVE PERFORMANCE COMPARISON UNDER EXTREME WEATHER SCENARIOS

Evaluation Metric	THRC	MARL	Proposed
Lifecycle energy consumption (10^4 kWh)	120	111	105
Peak water level under extreme rain (m)	5.51	4.89	4.30
Limit-exceeding alarm frequency (times)	15	8	1
Preventive action lead time (h)	0	1.5	4.0

Energy Efficiency Optimization: Under regular scheduling scenarios, the proposed architecture significantly outperformed

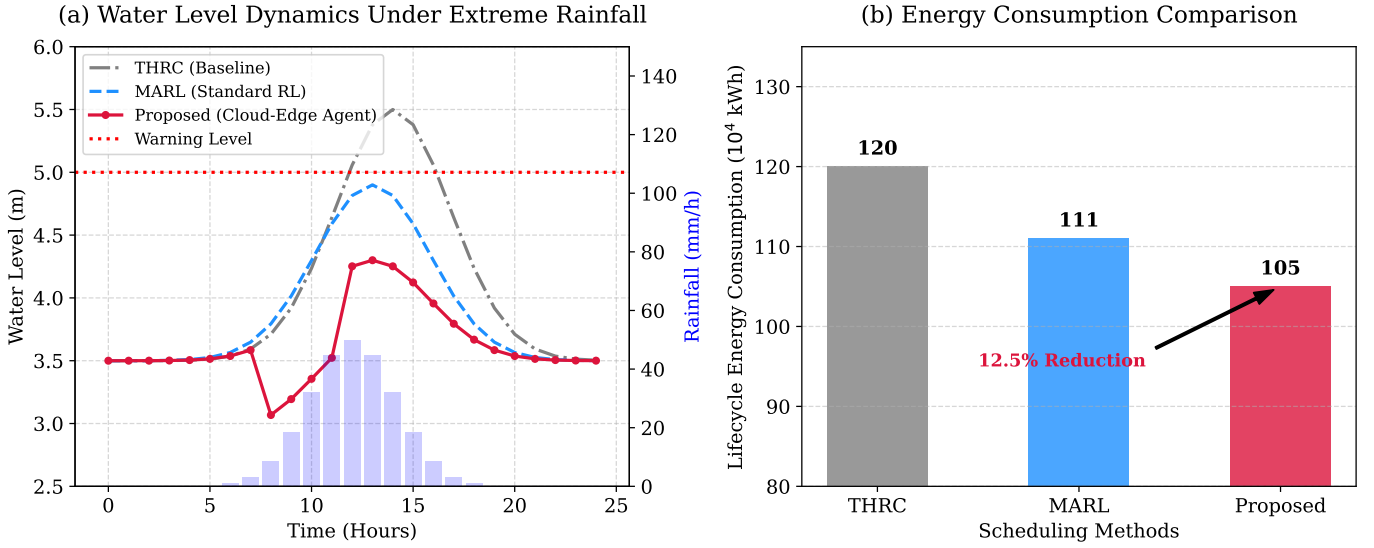


Fig. 3. Performance evaluation of the proposed architecture. (a) Real-time water level control dynamics under simulated extreme rainfall. The proposed method proactively lowers the water level to preserve storage capacity, avoiding the warning limit. (b) Comparison of lifecycle energy consumption, demonstrating a 12.5% reduction achieved by the proposed method.

the baselines. By leveraging the LLM’s CoT reasoning and the DP baseline’s optimization of Time-of-Use (ToU) electricity pricing, the multi-agent system effectively executed collaborative water storage during off-peak hours and smooth unit switching. Experimental results demonstrated a 12.5% reduction in the overall lifecycle energy consumption of the pumping station cluster compared to THRC.

Safety, Robustness, and Extreme Weather Response:

The system’s robustness was rigorously tested under simulated extreme rainfall and sudden equipment failure conditions. Benefiting from the RAG-enhanced domain knowledge base and the DP-RL hybrid fine-tuning, the proposed architecture successfully identified impending risks and initiated preventive water transfer 2–4 hours in advance. Consequently, the frequency of critical water-level limit-exceeding alarms was reduced by over 80% compared to the standard MARL approach, proving the system’s superior capability in bridging the gap between semantic risk cognition and physical execution.

V. CONCLUSION

In this paper, we proposed a novel cloud-edge hybrid intelligent scheduling framework for large-scale pumping station clusters, aiming to overcome the inherent limitations of traditional static rules and pure model-free reinforcement learning. By seamlessly integrating an advanced FastAPI-based LLM proxy with dynamic Model Context Protocol (MCP) tool execution, the system establishes a robust bridge between high-level semantic reasoning and low-level physical control. The introduction of RAG-enhanced domain knowledge successfully suppresses the “hallucination” phenomenon common in general LLMs, while the DP-RL hybrid optimization guarantees both strict safety boundaries and high dynamic adaptability. Extensive simulations demonstrate that the

proposed architecture significantly improves energy efficiency and effectively mitigates urban waterlogging risks by initiating preventative scheduling during extreme weather events. Future work will focus on scaling this multi-agent collaborative framework to broader regional water grids and integrating more granular multi-modal meteorological forecasting models.

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